
1. What is DAX? Why is it used?

DAX (Data Analysis Expressions) is a **formula language** used in **Power BI**, **Power Pivot**, and **Analysis Services Tabular Models** to perform data analysis and calculations.

◆ Purpose of DAX:

- To create **custom calculations** and **aggregations** in Power BI models.
- To build **calculated columns**, **measures**, and **calculated tables**.
- To perform **data modeling** and **advanced analytics**, such as time intelligence, filtering, and ranking.

💡 In short:

DAX enhances raw data by adding analytical logic that helps users generate insights beyond basic reporting.

2. What are the data types supported by DAX?

DAX supports the following **core data types**:

Data Type	Description	Example
Numeric	Includes integers, decimals, and currency values.	123, 45.67
Boolean	Represents TRUE or FALSE values.	TRUE(), FALSE()
String (Text)	Represents textual data.	"Power BI"
Date/Time	Stores dates and time values.	DATE(2025,11,2)
Currency	Fixed-point decimal data type with four decimal digits of precision.	\$123.4567
Blank (Null)	Represents missing or empty values.	BLANK()

3. What is the difference between a calculated column and a measure?

Feature	Calculated Column	Measure
Definition	Computed row by row and stored in the table.	Computed on demand during visualization or aggregation.
Storage	Stored in the model (increases model size).	Not stored — calculated at runtime (more efficient).
Evaluation Context	Uses row context (calculated per row).	Uses filter context (depends on applied filters).
Example	Profit = Sales[Revenue] - Sales[Cost]	Total Profit = SUM(Sales[Revenue]) - SUM(Sales[Cost])
Usage	Useful for intermediate calculations or row-level data.	Useful for aggregations and reporting KPIs.

💡 Tip:

Use **calculated columns** only when necessary — prefer **measures** for performance and efficiency.

4. How to exclude all filters while doing calculations on any measure?

To ignore all filters, you can use the **ALL()** function.

Syntax:

```
CALCULATE([Measure], ALL(Table))
```

Example:

```
Total Sales (Ignore Filters) = CALCULATE(SUM(Sales[Amount]), ALL(Sales))
```

✓ This formula calculates total sales **ignoring all filters** (like date, region, or product filters) applied in the report visuals.

If you want to ignore filters from specific columns only:

```
CALCULATE([Measure], ALL(Sales[Region]))
```

5. What are circular dependencies? How can you avoid them in DAX?

Definition:

A **circular dependency** occurs when two or more calculated elements (columns, measures, or tables) depend on each other **directly or indirectly**, creating a calculation loop.

Example:

- Column A depends on Column B
- Column B depends on Column A

This causes Power BI to throw a **circular dependency error**.

How to avoid circular dependencies:

1. **Break the dependency chain** — ensure one calculation does not depend on the result of another that refers back to it.
 2. **Separate calculations** into independent measures or tables.
 3. **Use variables (VAR)** to structure your expressions clearly.
 4. **Use measures instead of calculated columns** where possible (since measures are evaluated contextually, not stored).
 5. **Recheck relationships** in your data model — redundant relationships can also trigger circular references.
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